NPTEL Online Certification Course
Fundamentals of Artificial Intelligence
MOOCs; July - Dec 2026
Indian Institute of Technology Guwahati
Assignment No. 5: Detailed Solution

10 Marks

Each question carries 01 Mark each.

Q1. Which ONE of the following best describes Declarative Semantics in First Order Logic?

A. It specifies the order in which inference rules are applied.
B. It associates symbols in the language with objects, functions and relations of a conceptualization to determine the truth of sentences.
C. It converts quantified sentences into propositional logic.
D. It defines the syntax of well-formed formulae.
Correct Answer: (B)

Q2. Consider the following statements regarding conceptualization:

I. Every conceptualization contains a universe of discourse.
II. Every conceptualization specifies concepts and relationships among the objects in the domain.

Choose the correct option.

A. I only
B. II only
C. Both I and II
D. Neither I nor II
Correct Answer: (C)

Q3. Facts and rules that attempt to capture the important facts and concepts about a domain are collectively referred to as:

A. Interpretations
B. Axioms
C. Models
D. Proofs
Correct Answer: (B)

Q4. Which inference rule is NOT a sound deductive rule in classical first-order logic?

A. Modus Ponens
B. Modus Tollens
C. Resolution
D. Abduction
Correct Answer: (D)

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Q5. Which inference step is illustrated below?

$\forall x$ Likes(x, IceCream). Therefore Likes(Rahul, IceCream).

A. Existential Generalization
B. Universal Instantiation
C. Existential Instantiation
D. Resolution

Correct Answer: (B)

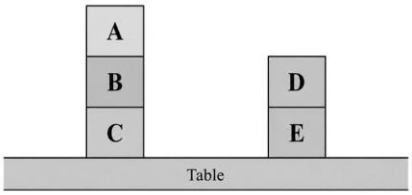
For Q6 – Q10

Consider the following Blocks World:
The predicates have the following meanings:

- on(x, y) : Block x is immediately on top of block y.
- above(x, y) : Block x is somewhere above block y (not necessarily immediately).
- clear(x) : No block is on top of block x.
- table(x) : Block x is directly on the table.

The knowledge base consists of the following facts:

- on(A, B)
- on(B, C)
- on(D, E)
- table(C)
- table(E)
- clear(A)
- clear(D)



The following rules are also known to be true.

- $\forall x \forall y$ (on(x, y) $\rightarrow$ above(x, y))
- $\forall x \forall y \forall z$ ((above(x, y) $\wedge$ above(y, z)) $\rightarrow$ above(x, z))
- $\forall x$ (clear(x) $\rightarrow$ $\neg \exists y$ on(y, x))

Q6. Which fact can be inferred using a single application of Modus Ponens?

A. above(A, C)
B. above(A, B)
C. clear(C)
D. table(A)

Correct Answer: (B)

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Q7. Which relation requires applying the transitivity rule?

A. above(A, B)
B. above(B, C)
C. above(A, C)
D. table(C)
Correct Answer: (C)

Q8. If the fact on(B,C) is removed, which inferred relation can no longer be proved?

A. above(D, E)
B. above(A, B)
C. above(A, C)
D. clear(A)
Correct Answer: (C)

Q9. Which statement is NOT entailed by the given knowledge base?

A. above(D, E)
B. ¬∃y on(y, A)
C. ¬∃y above(y, A)
D. above(A, C)
Correct Answer: (C)

Q10. Which of the following propositions is logically entailed by the knowledge base but is neither explicitly given as a fact nor derivable using a single application of Modus Ponens?

A. above(A, B)
B. above(D, E)
C. ¬∃y on(y, A)
D. above(A, C)

Correct Answer: (D)

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